



ICO HANGAR HACKATHON 2024

RepoRevive

AI-powered abandoned repository analysis and revival assistant. Because great code deserves a second chance.



Powered by Claude AI



GitHub API Integration



React + Node.js



Docker Ready

Team: icohangar-ops | Track: Developer Tools & Open Source Sustainability | v2.0

Open Source is Dying — One Abandoned Repo at a Time

2.1M+

ABANDONED GITHUB REPOS

73%

HAVE REVIVAL POTENTIAL

\$47B

VALUE OF ABANDONED OSS

0

GOOD TOOLS TO REVIVE THEM

1 No visibility into what's worth saving

Maintainers have no data-driven way to assess whether an abandoned repo is worth their time and effort to revive.

2 No clear path from "abandoned" to "active"

Even motivated maintainers don't know where to start. Documentation? CI? Community? The order matters and no one tells you.

3 Community re-engagement is hard

Former contributors have moved on. Attracting new ones requires the right issues, the right README, the right strategy.

RepoRevive: From Abandoned to Active in <5 Minutes

One URL. Full AI-powered diagnosis and revival roadmap — specific to your repository's code, community, and history.



Staleness Scoring

6-dimension algorithm: commit frequency, issue age, release recency, contributor count, stars, and archive status.



Revival Potential Score

Data-driven assessment of whether a repo is worth reviving based on community signals.



AI Revival Plans

Claude Opus generates quick wins, short-term goals, long-term roadmap, and community strategy tailored to your repo.



README Generator

Professional README.md auto-generated from your actual codebase, languages, and project context.



Good First Issues

AI-suggested beginner-friendly issues to attract new contributors and re-engage the community.



Revival Checklist

Interactive checklist covering docs, CI/CD, legal, community, and releases — with effort estimates.

HOW IT WORKS

Three Steps to Revive Any Repository



1. Analyze

Paste a GitHub URL. We fetch commits, issues, contributors, languages, and releases via the GitHub API.



2. Score

Our algorithm calculates Staleness (0–100) and Revival Potential (0–100) with a health grade A–F.



3. Revive

Claude AI generates a fully personalized revival plan, README, good-first issues, and interactive checklist.

72

Staleness Score

No commits in 14 months, 23 stale issues, no releases

68

Revival Potential

847 stars, 134 forks, 8 contributors — worth saving!

TECHNOLOGY

Modern, Production-Ready Stack

FRONTEND

React 18

React Router v6

TanStack Query

Recharts

Axios

CSS Variables

AI / INTELLIGENCE

Anthropic Claude Opus

@anthropic-ai/sdk

Prompt Engineering

Structured JSON Output

BACKEND

Node.js 20

Express.js

@octokit/rest

node-cache

Helmet

Rate Limiting

DEVOPS & INFRASTRUCTURE

Docker

GitHub Actions

Jest

nginx

Vercel

Railway

IMPROVEMENTS MADE

v1.0 → v2.0: A Complete Rebuild

Feature	v1.0 (Original)	v2.0 (This Hackathon)
Staleness detection	Basic date check only	6-dimension weighted algorithm
AI analysis	None	Claude Opus revival plans
README generator	None	AI-powered, repo-specific
Good First Issues	None	AI-generated suggestions
Revival checklist	None	Interactive + effort estimates
Language breakdown	None	Visual percentage chart
Security	None	Helmet, rate limiting, CORS
Tests	None	Unit + integration (Jest)
CI/CD	None	GitHub Actions pipeline
Docker support	None	Full docker-compose setup

A Multi-Billion Dollar Problem in Open Source

420M+

TOTAL GITHUB REPOSITORIES

2.1M+

ABANDONED (>1YR NO COMMITS)

\$8.8B

OSS CONTRIBUTION VALUE (ANNUAL)



Enterprise Users

Engineering teams that depend on abandoned OSS dependencies and need to fork + revive them for long-term support.



Individual Maintainers

Developers returning to side projects after months away, or inheriting ownership of unmaintained libraries.



OSS Foundations

Apache, Linux Foundation, CNCF — organizations responsible for maintaining hundreds of sub-projects.



Student Contributors

Hacktoberfest participants and students looking for projects to contribute to that actually need help.

PRODUCT DEMO

See RepoRevive in Action

```
● ● ● POST /api/analyze

$ curl -X POST https://reporevive-api.railway.app/api/analyze \
  -H "Content-Type: application/json" \
  -d '{"repo": "nicehash/NiceHashQuickMiner"}'

// Response (truncated)
{
  "scores": { "staleness": 74, "revivalPotential": 61, "healthGrade": "C" },
  "revivalPlan": {
    "summary": "NiceHashQuickMiner was a popular mining tool with strong community traction...",
    "quickWins": ["Update README with current status", "Label stale issues", ...]
  }
}
```

 Demo Video: youtu.be/reporevive-demo

 Live App: reporevive.vercel.app

Key Learnings & Next Steps

KEY LEARNINGS

1 **GitHub API rate limits are real**

Caching is essential — we implemented 5-min TTL to handle unauthenticated bursts gracefully.

2 **Structured AI output requires discipline**

Claude responds better with explicit JSON schema in the prompt plus fallback parsing.

3 **Staleness ≠ worthless**

Some of the most valuable repos are the most abandoned — stars and forks tell a different story.

NEXT STEPS

Q1 **GitHub App Integration**

Install RepoRevive as a GitHub App for automated monitoring and weekly revival reports.

Q2 **Batch Analysis & Discovery**

Analyze entire GitHub organizations or search for abandoned repos by topic/language globally.

Q3 **Revival Marketplace**

Connect abandoned repo owners with developers who want to take over maintenance.

Q4 **Enterprise & SaaS Tier**

Team dashboards, Slack notifications, Jira integration, and private repo support.

[GET STARTED](#)

Ready to Revive Something?

Every abandoned repository has a story.
RepoRevive helps you write the next chapter.

MIT

OPEN SOURCE LICENSE

<5 min

FROM URL TO REVIVAL PLAN

Free

SELF-HOSTED FOREVER

Try RepoRevive Nowgithub.com/icohangar-ops/reporevive

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